Introduction to Economics

Class 14 — Monopolistic competition; problems on Blocks 3–4

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Universitat de Barcelona · Group A6

12 November 2026

Quiz on Problem Set 3

15 minutes · pens down at 10:15

Use only your copy of Problem Set 3 and your own worked solutions.

Write your name and your copy number ("Copy NN", top right of your problem set).

Answers as numbers in the boxes; use the frames for your working.

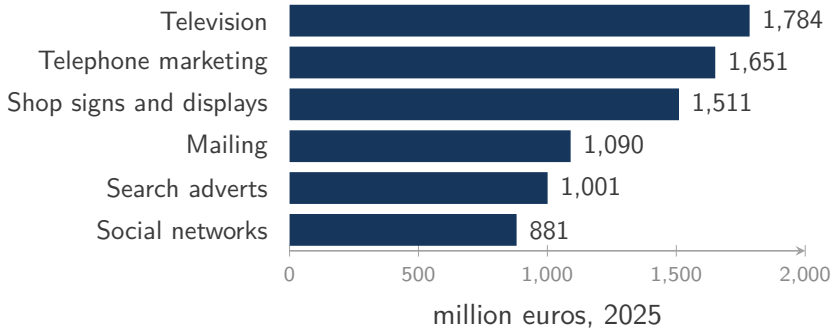
Questions on both sides. Hand in the sheet; keep your problem set.

Plan for today

- Monopolistic competition: many sellers, each a little different
- A coworking space in the short run and the long run
- Why firms advertise
- Practice on the grid: a price taker's costs and profit
- Two curves from a table: a shift, the elasticity; a price cap

Spain's advertising: 12.7 billion euros in 2025

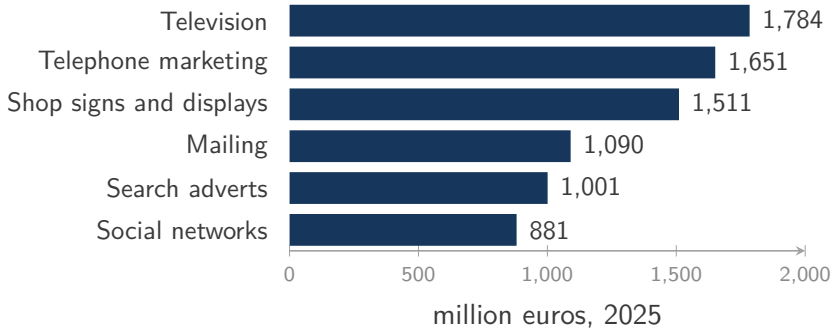
the six largest of 24 channels



InfoAdex, Estudio de la Inversión Publicitaria en España 2026 (2025 data): total 12,745.4 million euros, 0.76 % of gross domestic product (GDP); per person: ÷ about 49.6 million residents (Instituto Nacional de Estadística).

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About 260 euros for every person living in Spain.

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Monopolistic competition: many sellers, each a little different, and anyone can open one

	Perfect competition (class 12)	Monopolistic competition (today)	Monopoly (class 13)
Sellers	many		one
Product	identical		no close substitute
Entry	free		blocked
One firm's demand	flat at the price		the whole market's

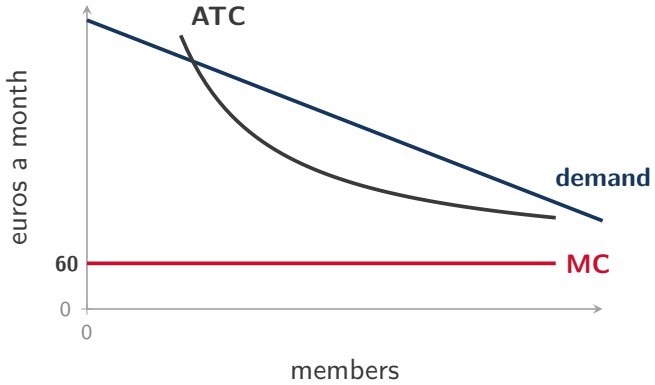
Price taker: sells all it wants at the market price (class 12, slide 6) · monopoly: the only seller, no close substitute (class 12, slide 20).

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Sellers	many	many	one
Product	identical	each a little different	no close substitute
Entry	free	free	blocked
One firm's demand	flat at the price	slopes down	the whole market's

Price taker: sells all it wants at the market price (class 12, slide 6) · monopoly: the only seller, no close substitute (class 12, slide 20).

In the short run a coworking space acts like a monopoly: 40 members at 220 euros, a profit of 2,800



60 desks

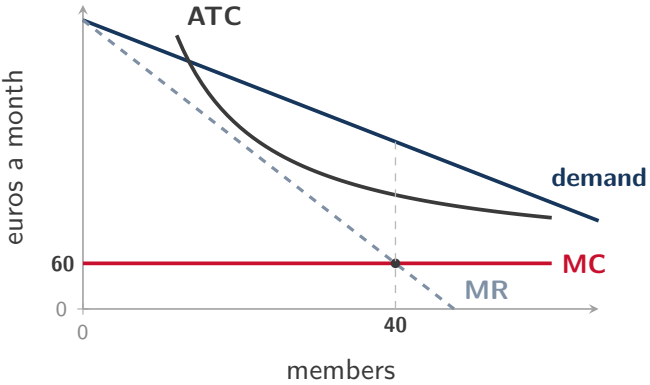
rent, furniture and the owner's time: 3,600 a month

each member costs 60 a month

Marginal revenue (MR): what one more member adds to revenue (class 13, slide 4).

Average total cost (ATC): total cost ÷ members (class 11, slide 10) · MC: marginal cost.

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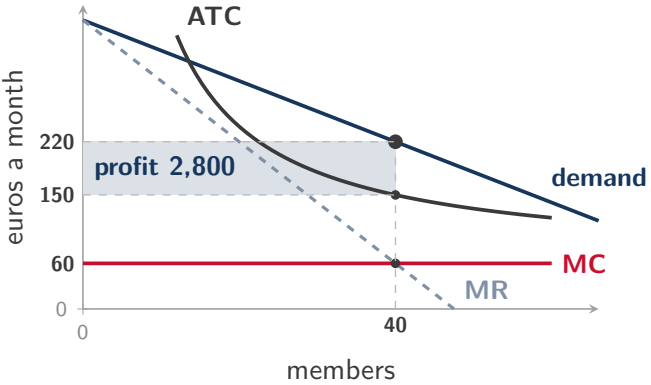
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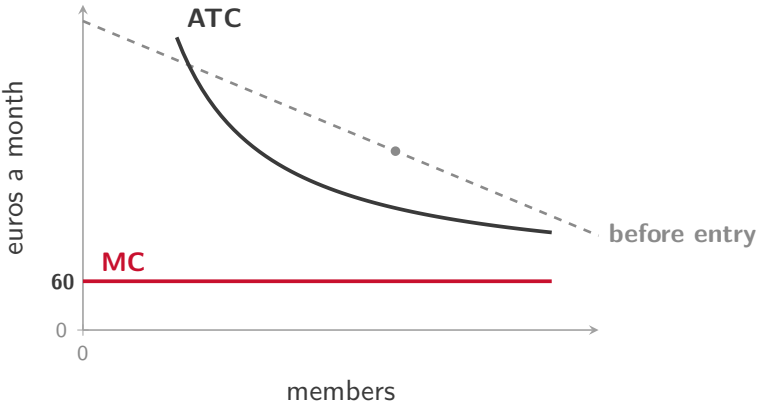
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Like a monopoly:
sell while MR beats MC,
read the price off demand.

Marginal revenue (MR): what one more member adds to revenue (class 13, slide 4).

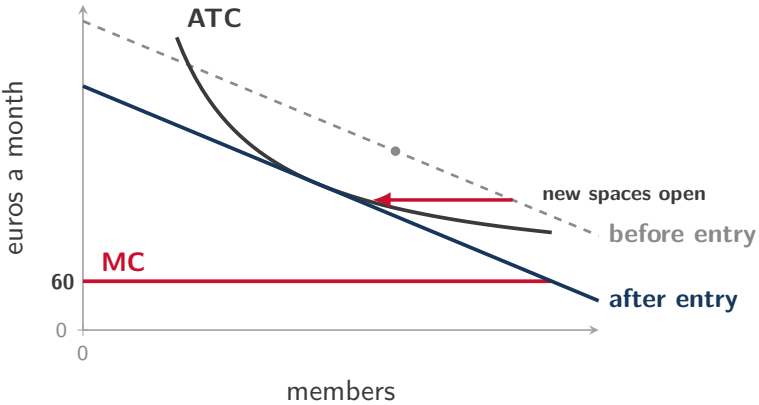
Average total cost (ATC): total cost ÷ members (class 11, slide 10) · MC: marginal cost.

New spaces open: each one's demand shifts left until profit is zero — 30 members at 180



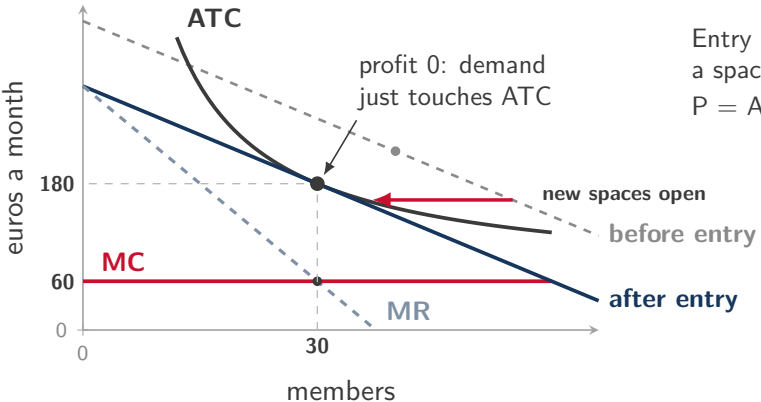
Profit here is economic profit: every cost counted, the owner's own time included (class 11, slide 6).

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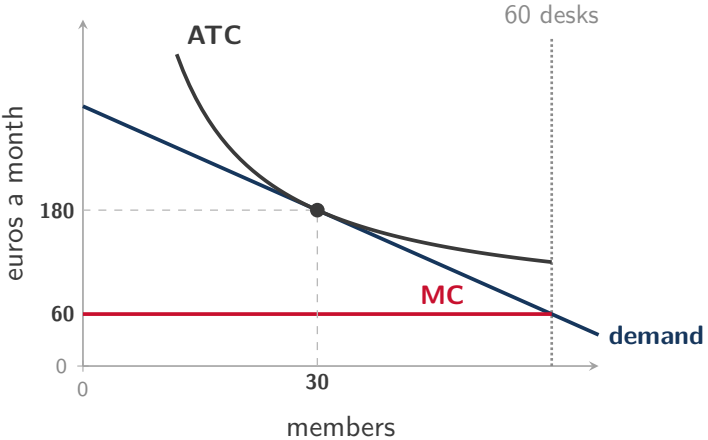
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Entry stops when the best a space can do is zero profit: P = ATC.

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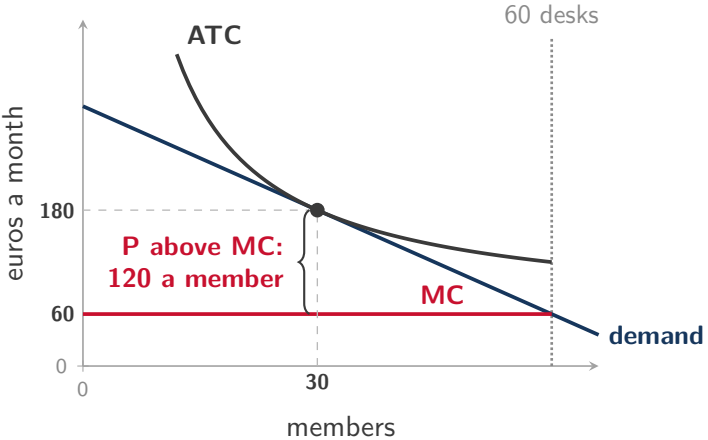
Zero profit, yet the price is above MC and desks stand empty



Perfect competition, long run: price = lowest ATC (class 12, slide 18).

Fixed cost spread thinner, ATC falls (class 11, slide 15).

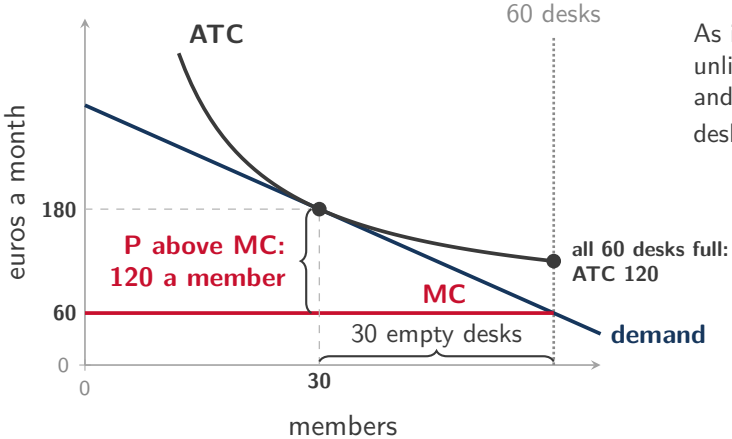
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Zero profit, yet the price is above MC and desks stand empty



As in class 12, P = ATC; unlike class 12, P > MC and ATC is still falling: desks stand empty.

Perfect competition, long run: price = lowest ATC (class 12, slide 18).

Fixed cost spread thinner, ATC falls (class 11, slide 15).

With the price above MC, each extra member adds 120: so the space advertises

One more member: pays 180, costs 60 → adds 120 a month

A price taker: one more buyer at P = MC adds nothing

A price taker produces while MC is below the price (class 12, slide 8) · 12.7 billion euros a year (slide 3).

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Critics

- shapes tastes, says little
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Defenders

- tells prices, products, places
- helps new firms enter
- a costly campaign signals quality

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Monopolistic competition: many sellers, each a little different, free entry.

Short run: like a monopoly.
Long run: entry until profit is zero.

Even then the price stays above MC, and capacity is left unused.

Above MC, one more customer pays: that is why firms advertise.

Break

Back at 11:00

A pomegranate farm's costs: the table — four minutes alone

A farm sells crates of pomegranates at the market price of 30 euros a crate; it is a price taker. The table gives its total cost (TC) for 0 to 5 crates a day. [5.5 points; table cells as marked]

(a) Complete the table: the marginal cost (MC) of each crate (the extra cost of that crate) and the average total cost (ATC) at 2, 3 and 4 crates.

Crates a day	0	1	2	3	4	5
Total cost, TC (euros)	24	34	44	54	80	114
MC of that crate (euros) [0.5]	–					
ATC (euros a crate) [1]	–	–				–

TC: total cost · MC: marginal cost, the extra cost of one more crate · ATC: average total cost, TC ÷ crates.

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MC of that crate (euros) [0.5]	–	**10**	**10**	**10**	**26**	**34**
ATC (euros a crate) [1]	–	–				–

MC = change in TC: 34 − 24 = 10, 44 − 34 = 10, 54 − 44 = 10, 80 − 54 = 26, 114 − 80 = 34

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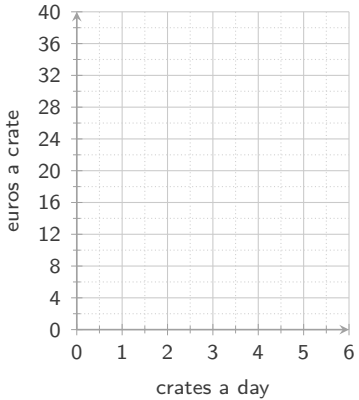
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The farm on the grid: MC, ATC and profit — four minutes alone

The same farm, at the market price of 30 euros a crate. Its costs (euros):



[10.5 points; drawing 4.5; boxes as marked]

crates a day	0	1	2	3	4	5
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MC	–	10	10	10	26	34
ATC	–	–	22	18	20	–

(b) Draw MC as steps: over each crate, a horizontal line at that crate's MC. Mark the ATC at 2, 3 and 4 crates with dots and join them. Draw the price as a line labelled P.

(c) Best quantity (crates a day) [2] Profit (euros a day) [2]

Shade the profit as a rectangle; write profit inside it.

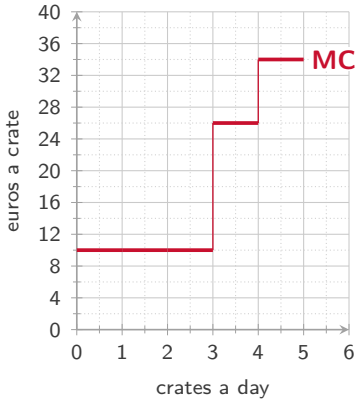
(d) At 12 euros a crate: produce or shut down in the short run? Circle one; the reason in one line. [2]

(Produce) (Shut down)

MC: marginal cost · ATC: average total cost · AVC: average variable cost, variable cost ÷ crates · P: price.

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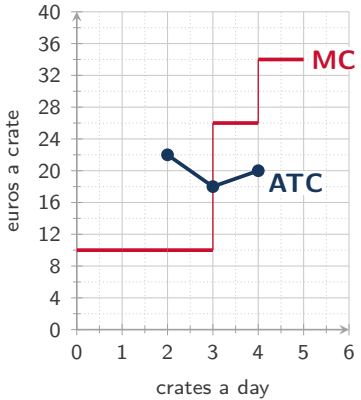
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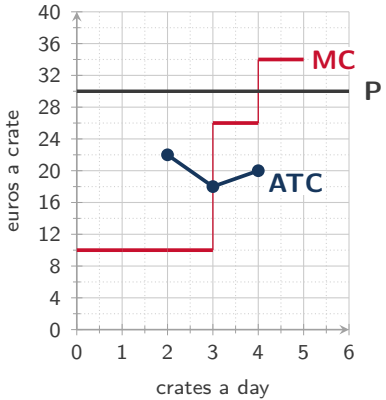
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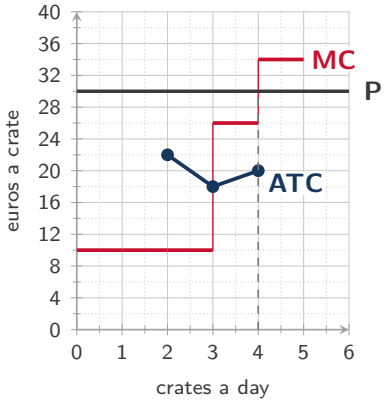
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4: the 5th costs 34

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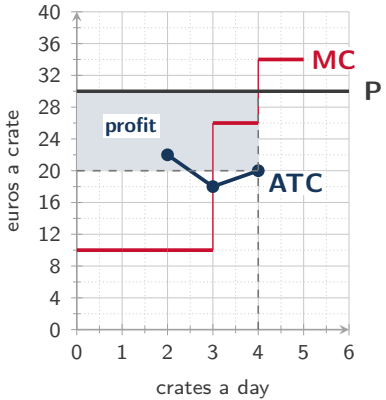
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4: the 5th costs 34

$(30 - 20) \times 4 = 40$

Shade the profit as a rectangle; write profit inside it.

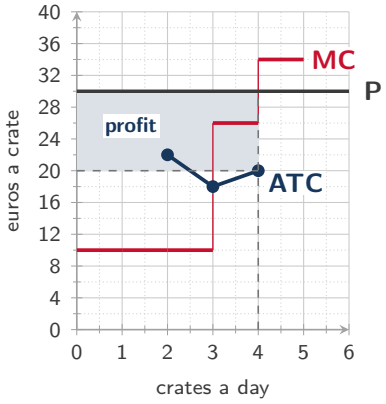
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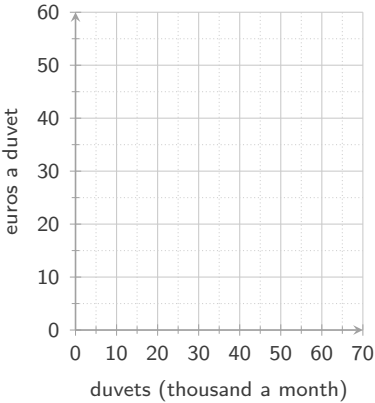
Produce / Shut down

12 is above the lowest AVC, 10 (1–3 crates): it loses 18, not the fixed 24

MC: marginal cost · ATC: average total cost · AVC: average variable cost, variable cost ÷ crates · P: price.

Duvets: two curves from a table, then a shift — six minutes alone

The market for duvets (quantities in thousand duvets a month). The table gives the quantities demanded and supplied at three prices; both curves are straight lines.



[12 points; drawing 6; boxes as marked]

Price (euros a duvet)	25	35	45
Quantity demanded (thousand a month)	60	40	20
Quantity supplied (thousand a month)	15	25	35

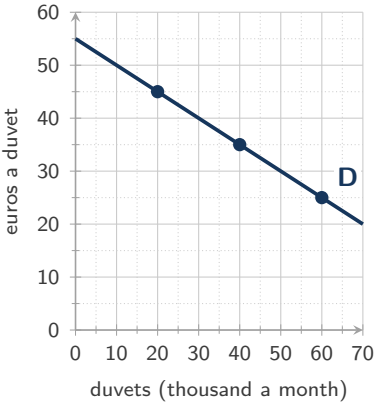
(a) Draw the demand curve and the supply curve through the table's points, label them D and S, and mark the equilibrium E_1.

(b) The price of polyester, an input used to make duvets, falls. At every price, sellers now offer 30 thousand more duvets a month than before. Draw the new supply curve, label it S_2, and mark the new equilibrium E_2.

	Price (euros) [1.5]	Quantity (thousand) [1.5]
(a) Before: E_1		
(b) After: E_2		

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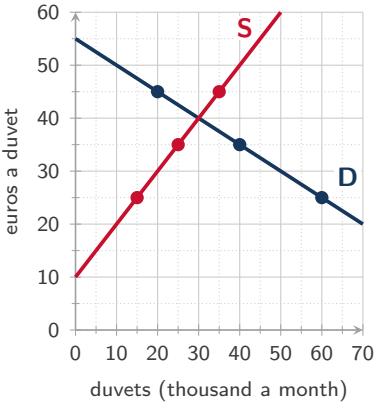
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Quantity demanded (thousand a month)	60	40	20
Quantity supplied (thousand a month)	15	25	35

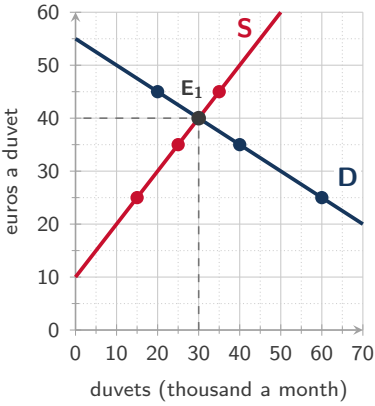
(a) Draw the demand curve and the supply curve through the table's points, label them D and S, and mark the equilibrium E_1.

(b) The price of polyester, an input used to make duvets, falls. At every price, sellers now offer 30 thousand more duvets a month than before. Draw the new supply curve, label it S_2, and mark the new equilibrium E_2.

	Price (euros) [1.5]	Quantity (thousand) [1.5]
(a) Before: E_1		
(b) After: E_2		

Duvets: two curves from a table, then a shift — six minutes alone

The market for duvets (quantities in thousand duvets a month). The table gives the quantities demanded and supplied at three prices; both curves are straight lines.



[12 points; drawing 6; boxes as marked]

Price (euros a duvet)	25	35	45
Quantity demanded (thousand a month)	60	40	20
Quantity supplied (thousand a month)	15	25	35

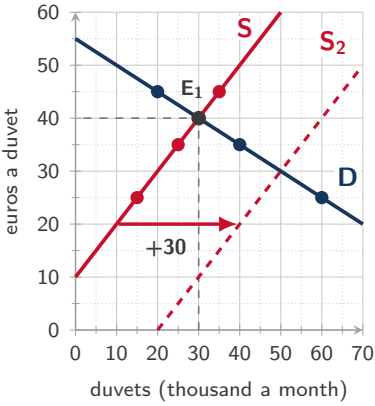
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	Price (euros) [1.5]	Quantity (thousand) [1.5]
(a) Before: E_1	**40 euros**	**30 thousand**
(b) After: E_2		

Duvets: two curves from a table, then a shift — six minutes alone

The market for duvets (quantities in thousand duvets a month). The table gives the quantities demanded and supplied at three prices; both curves are straight lines.



[12 points; drawing 6; boxes as marked]

Price (euros a duvet)	25	35	45
Quantity demanded (thousand a month)	60	40	20
Quantity supplied (thousand a month)	15	25	35

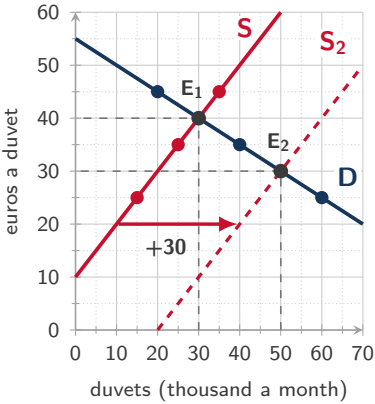
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	Price (euros) [1.5]	Quantity (thousand) [1.5]
(a) Before: E_1	**40 euros**	**30 thousand**
(b) After: E_2		

Duvets: two curves from a table, then a shift — six minutes alone

The market for duvets (quantities in thousand duvets a month). The table gives the quantities demanded and supplied at three prices; both curves are straight lines.



[12 points; drawing 6; boxes as marked]

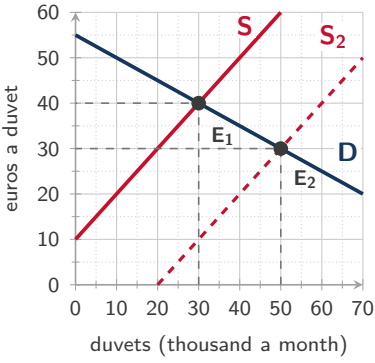
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Quantity demanded (thousand a month)	60	40	20
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(a) Draw the demand curve and the supply curve through the table's points, label them D and S, and mark the equilibrium E_1.

(b) The price of polyester, an input used to make duvets, falls. At every price, sellers now offer 30 thousand more duvets a month than before. Draw the new supply curve, label it S_2, and mark the new equilibrium E_2.

	Price (euros) [1.5]	Quantity (thousand) [1.5]
(a) Before: E_1	**40 euros**	**30 thousand**
(b) After: E_2	**30 euros**	**50 thousand**

The duvets: the elasticity from E_1 to E_2 — two minutes alone



[2 points; working shown]

E_1: 40 euros, 30 thousand · E_2: 30 euros, 50 thousand

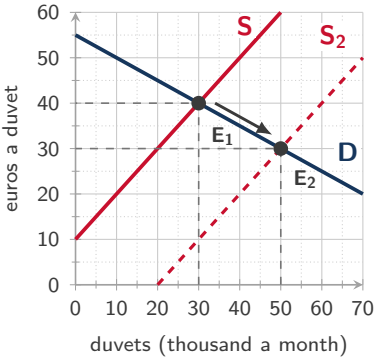
(c) Demand did not move, so E_1 and E_2 are two points on the demand curve. Compute the price elasticity of demand between them by the midpoint method, and write it in the box as a positive number (a fraction or a decimal).

Elasticity (a positive number) [2]

Working:

E_1, E_2: the equilibrium before and after · price elasticity of demand: midpoint method, as a positive number.

The duvets: the elasticity from E_1 to E_2 — two minutes alone



[2 points; working shown]

E_1: 40 euros, 30 thousand · E_2: 30 euros, 50 thousand

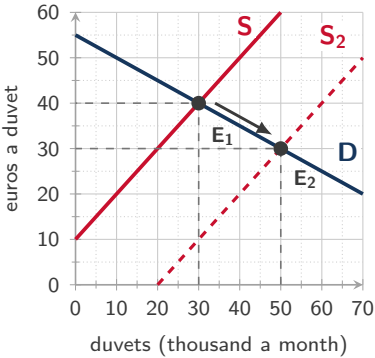
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Elasticity (a positive number) [2]

Working: **Quantity: $50 - 30 = 20$; average $(30 + 50) \div 2 = 40$; $20 \div 40 = 1/2$**

E_1, E_2: the equilibrium before and after · price elasticity of demand: midpoint method, as a positive number.

The duvets: the elasticity from E_1 to E_2 — two minutes alone



[2 points; working shown]

E_1: 40 euros, 30 thousand · E_2: 30 euros, 50 thousand

(c) Demand did not move, so E_1 and E_2 are two points on the demand curve. Compute the price elasticity of demand between them by the midpoint method, and write it in the box as a positive number (a fraction or a decimal).

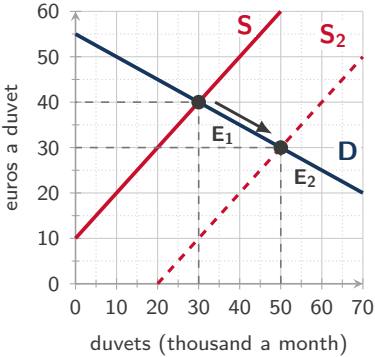
Elasticity (a positive number) [2]

Working: **Quantity: $50 - 30 = 20$; average $(30 + 50) \div 2 = 40$; $20 \div 40 = 1/2$**

Price: $40 - 30 = 10$; average $(40 + 30) \div 2 = 35$; $10 \div 35 = 2/7$

E_1, E_2: the equilibrium before and after · price elasticity of demand: midpoint method, as a positive number.

The duvets: the elasticity from E_1 to E_2 — two minutes alone



[2 points; working shown]

E_1: 40 euros, 30 thousand · E_2: 30 euros, 50 thousand

(c) Demand did not move, so E_1 and E_2 are two points on the demand curve. Compute the price elasticity of demand between them by the midpoint method, and write it in the box as a positive number (a fraction or a decimal).

Elasticity (a positive number) [2]

7/4 = 1.75

Working: **Quantity: 50 − 30 = 20; average (30 + 50) ÷ 2 = 40; 20 ÷ 40 = 1/2**

Price: 40 − 30 = 10; average (40 + 30) ÷ 2 = 35; 10 ÷ 35 = 2/7

(1/2) ÷ (2/7) = 1/2 × 7/2 = 7/4 = 1.75: demand is elastic here

E_1, E_2: the equilibrium before and after · price elasticity of demand: midpoint method, as a positive number.

Prawns at Christmas: a price cap — five minutes alone

The grid shows a city's market for prawns in Christmas week (quantities in thousand kilos): demand D and supply S. The city sets a maximum price (a price cap) of 20 euros a kilo.

[9 points; drawing 2; boxes as marked]



(a) Draw the cap as a line and label it cap.

(b) The quantity supplied, the quantity demanded and the shortage at the cap (thousand kilos).

Supplied [1.5] Demanded [1.5] Shortage [1]

(c) Only the quantity supplied is sold, and it goes to the buyers who value it most. The deadweight loss (DWL) with the cap, in thousand euros; write DWL inside its area.

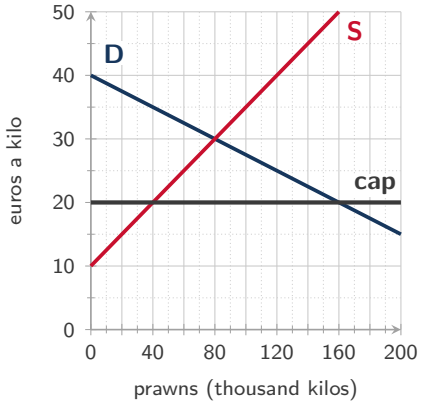
DWL [3]

Thousand kilos × euros a kilo = thousand euros. A triangle is ½ × base × height.

Prawns at Christmas: a price cap — five minutes alone

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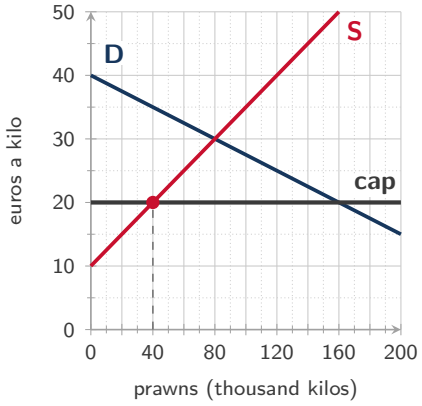
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[9 points; drawing 2; boxes as marked]



(a) Draw the cap as a line and label it cap.

(b) The quantity supplied, the quantity demanded and the shortage at the cap (thousand kilos).

Supplied [1.5] Demanded [1.5] Shortage [1]

40		

(c) Only the quantity supplied is sold, and it goes to the buyers who value it most. The deadweight loss (DWL) with the cap, in thousand euros; write DWL inside its area.

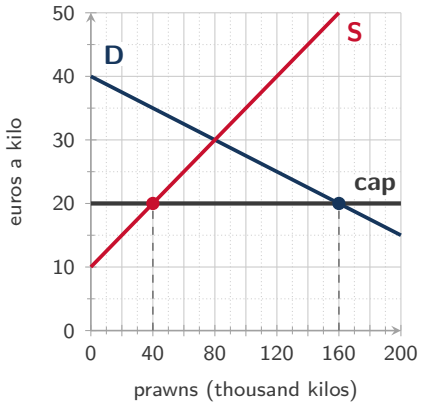
DWL [3]

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Prawns at Christmas: a price cap — five minutes alone

The grid shows a city's market for prawns in Christmas week (quantities in thousand kilos): demand D and supply S. The city sets a maximum price (a price cap) of 20 euros a kilo.

[9 points; drawing 2; boxes as marked]



(a) Draw the cap as a line and label it cap.

(b) The quantity supplied, the quantity demanded and the shortage at the cap (thousand kilos).

Supplied [1.5] Demanded [1.5] Shortage [1]

40	160	

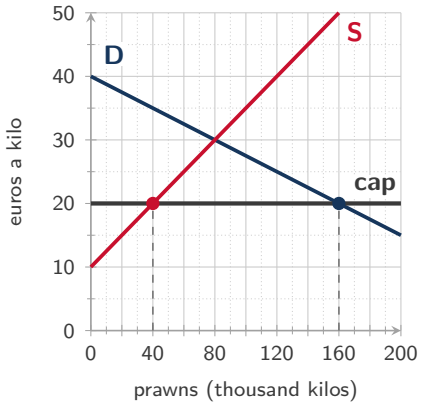
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DWL [3]

Thousand kilos × euros a kilo = thousand euros. A triangle is ½ × base × height.

Prawns at Christmas: a price cap — five minutes alone

The grid shows a city's market for prawns in Christmas week (quantities in thousand kilos): demand D and supply S. The city sets a maximum price (a price cap) of 20 euros a kilo.



[9 points; drawing 2; boxes as marked]

(a) Draw the cap as a line and label it cap.

(b) The quantity supplied, the quantity demanded and the shortage at the cap (thousand kilos).

Supplied [1.5] Demanded [1.5] Shortage [1]

40	160	160 − 40 = 120

(c) Only the quantity supplied is sold, and it goes to the buyers who value it most. The deadweight loss (DWL) with the cap, in thousand euros; write DWL inside its area.

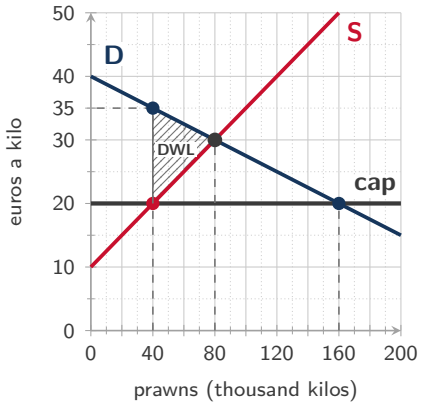
DWL [3]

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Prawns at Christmas: a price cap — five minutes alone

The grid shows a city's market for prawns in Christmas week (quantities in thousand kilos): demand D and supply S. The city sets a maximum price (a price cap) of 20 euros a kilo.

[9 points; drawing 2; boxes as marked]



(a) Draw the cap as a line and label it cap.

(b) The quantity supplied, the quantity demanded and the shortage at the cap (thousand kilos).

Supplied [1.5] Demanded [1.5] Shortage [1]

Supplied	Demanded	Shortage
40	160	160 − 40 = 120

(c) Only the quantity supplied is sold, and it goes to the buyers who value it most. The deadweight loss (DWL) with the cap, in thousand euros; write DWL inside its area.

DWL [3]

DWL
300

½ × (80 − 40) × (35 − 20) = 300 thousand euros

Thousand kilos × euros a kilo = thousand euros. A triangle is ½ × base × height.

Before Monday 16 November

- Monday 16 November: review for Midterm 1 (Blocks 1–4), the full two hours
- Thursday 19 November: Midterm 1 (Blocks 1–4), in class
- Practise in Sextant: the units the course page lists



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